

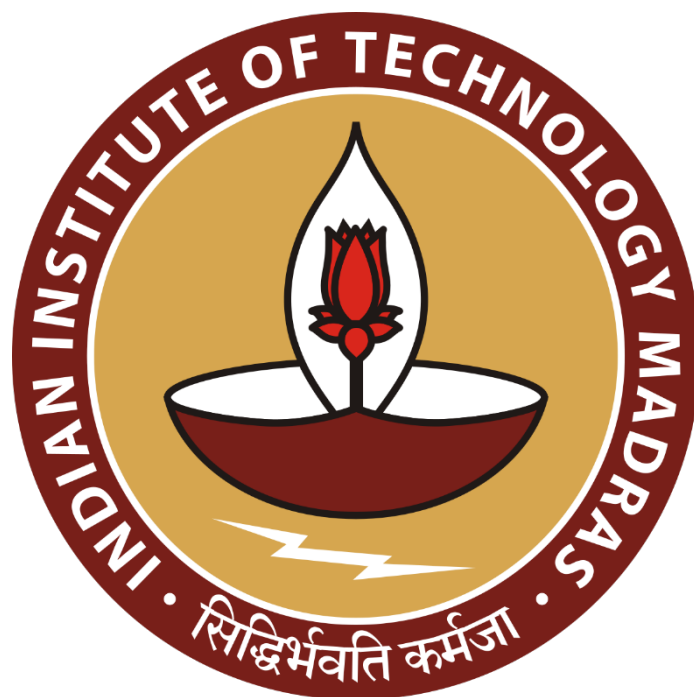
Optimal Inventory Management and Sales Forecasting for Janananma Pharmacy

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “Optimal Inventory Management and Sales Forecasting for Janananma Pharmacy”. I extend my appreciation to Janananma Medicals and surgicals, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate: 

Name: Nihila Pallath

Date: 02-07-2024

1 Executive Summary and Title

The project focuses on a small pharmacy located in Mannarkkad, Palakkad, Kerala. The business is B2C and deals in the segment of medical and surgical supplies, serving a diverse customer base in a bustling urban area.

This project aims to tackle the critical challenges faced by Janananma Pharmacy in managing its inventory and forecasting sales to ensure efficient stock management and minimize losses due to expired medicines.

The analysis proposed to begin with an examination of sales data, purchase data and inventory data to pinpoint stock management, brand diversity and expiry Management. Key challenges include fluctuating sales and market trends necessitating regular updates to maintain optimal inventory levels, the diversity of prescriptions from various doctors specifying brand names rather than chemical contents complicating inventory management and customers, and financial losses occurring due to medicines expiring before being sold makes impact on cash flow.

The recommendations from this project aim to enable Janananma Pharmacy to identify bottlenecks in their current inventory processes and provide practical solutions to improve stock management and sales forecasting. While these solutions will not resolve all issues, they offer targeted strategies to tackle key problems, ultimately contributing to better operational efficiency and financial health for the pharmacy.

2 Organization Background

Janananma Medicals and Surgicals, established in 2015 by Muhammed Ajmal in Mannarkkad, Palakkad, Kerala, has become a leading B2C healthcare business. Its success is rooted in a strong commitment to quality and customer satisfaction, efficiently managed through Remedy Plus Software. The company offers a wide range of healthcare products, including prescription drugs, health and wellness items, personal and home care products, baby care essentials, and medical supplies. This comprehensive approach ensures that the diverse needs of its customers are met.

The team at Janananma comprises five professionals: two with bachelor's degrees in pharmacy, two with degrees in accountancy, and one intern. They work on shift schedules to provide continuous service. With a strong focus on delivering high-quality products and exceptional, personalized care, Janananma Medicals and Surgicals aims to empower individuals to lead healthier, happier lives. The company's steady growth and increasing net worth highlight its pivotal role in the regional healthcare sector, making it a trusted name in the community.

3 Problem Statement

- 3.1 **Stock Management:** Maintaining the optimal level of inventory is crucial to ensure that the pharmacy does not run out of essential medicines while also avoiding overstocking, which can tie up capital and increase the risk of product expiration. The non availability forecast based stock keeping is one of the main problems faced by Janananma Medicals.
- 3.2 **Expiry Management:** Medicines reaching their expiry dates before being sold can lead to significant financial losses. Although companies may offer refunds for expired items, the invested capital is tied up for long periods without generating returns. And it will also increase the inventory cost as it needs some space to store all those things until it returns. Therefore, this leads to an increase in the cost of inventory.
- 3.3 **Brand Diversity:** The pharmacy is in a diverse area where customers receive prescriptions from various doctors who often specify brand names rather than the chemical content of the medicines. This necessitates stocking multiple brands of the same medicine, complicating inventory management.

4 Background of the Problem

The primary cause of the problems faced by Janananma Medicals and Surgicals is its Dynamic Market Challenges, Resource Allocation, Brand diversity complexity, Expiry management issues which hinder sales and revenue generation.

Mannarkkad is a diverse community where people from various cultural backgrounds reside. The local pharmacy faces challenges due to the impatience of customers who cannot wait a day or two for medication availability. Therefore, maintaining an adequate stock of medicines is essential.

The pharmacy often struggles with improper stock management, leading to many medicines expiring before they are sold, resulting in significant financial losses. Additionally, the demand for immediate availability of all prescribed medications drives the pharmacy to overstock, fearing the loss of customers.

This problem is further compounded by the large number of doctors in the area who prescribe medications based on brand names rather than their active ingredients. As a result, Janananma Medicals is compelled to stock the same medicine from different companies, complicating inventory management.

5 Problem Solving Approach

Methods Used with Justification:

- To ensure effective data analysis and practical problem-solving, I collected primary data on sales, purchases, and inventory from the pharmacy. The data cleaning and preparation process involved removing unnecessary columns and rows and replacing missing data using appropriate Excel functions. This ensures clean and consistent data, which is vital for accurate analysis, precise calculations, and reliable comparisons.
- Analysing the sales data to Implementing predictive analytics models to **forecast future demand** based on historical sales data and market trends. Accurate demand forecasting helps in maintaining optimal inventory levels, preventing both stock shortages and excess inventory. This method enhances the pharmacy's ability to meet customer needs efficiently.
- Developing an inventory rotation system and implementing expiry date tracking. This method helps in prioritizing the sale of products nearing their expiration date, thereby minimizing financial losses from expired stock and optimizing cash flow.
- Collecting historical prescription data to identify common brands and medicines prescribed by local doctors. By recognizing the patterns in prescriptions, the pharmacy can stock the brands that are most frequently recommended, improving customer satisfaction and ensuring a reliable supply of preferred medicines. Establishing a system to record and analyze customer requests helps in proper inventory management.

Intended Data Collection with Justification:

- Sales Data: It provides a clear picture of which products are in demand, helping to inform stock management and demand forecasting models.
- Purchase Data: It provides the expiry date for each product. And using the sales and purchase data together helping to understand demand of each product.
- Inventory Data: Current inventory levels, including details on stock quantities, is essential for tracking stock levels, and implementing an efficient inventory rotation system.

Analysis Tools with Justification:

- The primary tool for analysing the data is MS Excel. Data Cleaning and statistical analysis of sales, inventory data providing a good idea for strategic planning and optimization.
- Functions like VLOOKUP will help to organize and sort out the required data as a part of data cleaning, it helps in searching and retrieving data from a table based on a specific criterion.
- Pivot Table is used to summarize the large amounts of data, showing trends and patterns, such as total sales per product category or the inventory status over time.
- Regression from data analysis tool pack in excel used for the basic analysis

- Charts and Graphs like Pareto chart, Bar chart, Histogram and pie charts are used to visualize the data to identify trends and patterns.
- Forecasting Tools are used to forecast future sales trends and demand for specific products, aiding in better inventory planning and management.

6 Expected Timeline

6.1 Work Breakdown Structure:

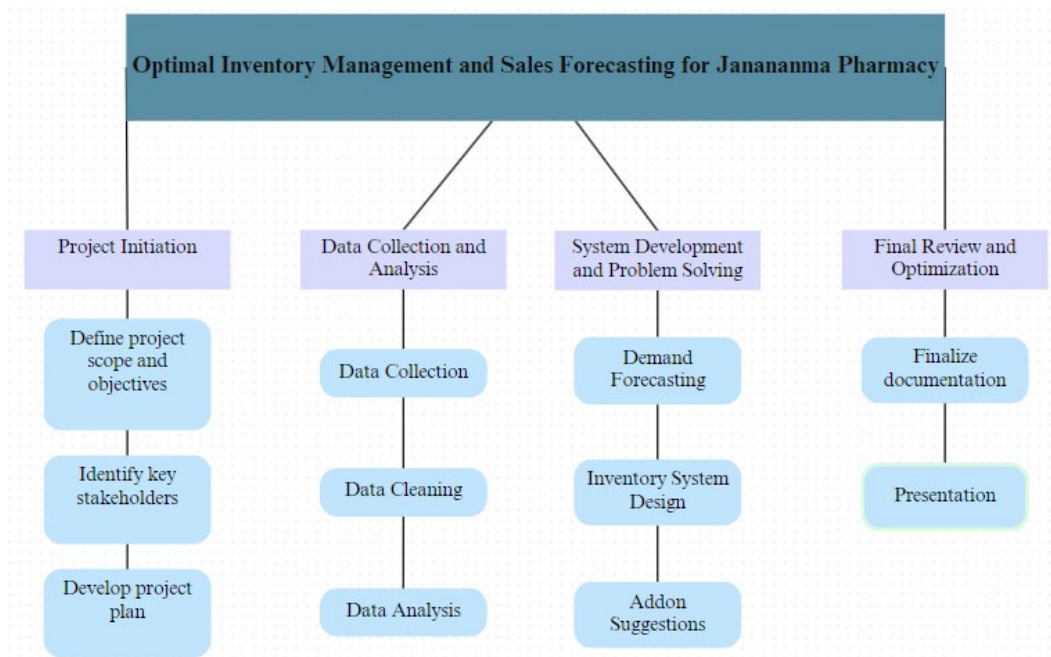


Figure 1: WBS for the Project

6.2 Gantt chart

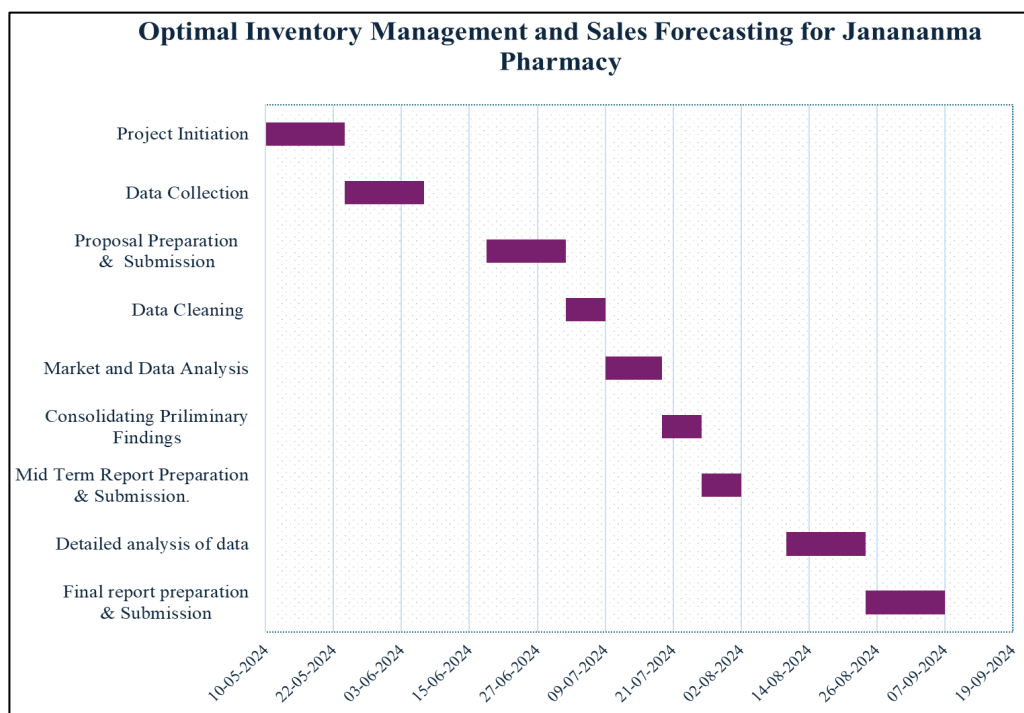


Figure 2: Expected timeline for completion of project.

7 Expected Outcome

7.1 Improved Inventory Management:

By accurately forecasting demand, the pharmacy will be able to maintain optimal stock levels, ensuring that essential medicines are always available, it will help to reduce stockouts. Effective demand forecasting and inventory management will prevent overstocking, reducing financial strain and waste associated with unsold inventory.

7.2 Enhanced Brand Diversity Handling:

By analyzing prescription patterns and customer preferences, the pharmacy can more effectively manage inventory for different brands, ensuring that the most popular and prescribed brands are always in stock. Meeting customer preferences for specific brands will lead to higher customer satisfaction and loyalty, potentially increasing sales and market share.

7.3 Effective Expiry Management:

Improved inventory rotation and demand forecasting will minimize the amount of stock that reaches its expiry date unsold, reducing direct financial losses. By avoiding the lock-up of capital in expired stock, the pharmacy will have better cash flow, allowing for more timely investments in new stock or other operational improvements.

The project will provide detailed documentation of all processes, methodologies, and systems implemented, ensuring sustainability and continuity. These outcomes will collectively help Janananma Medicals and Surgicals overcome its current challenges, improve its operational efficiency, and achieve better financial performance.